University of Massachusetts, Amherst
CICS
COMPSCI 683: Artificial Intelligence

Assignment Bonus #2: Overview Questions

Due: two weeks after the issue date in Gradescope

Important Instructions:

- Please upload your solution to Gradescope.
- You may work with a partner on your homework; however, if you choose to do so, **you must** write down the name of your partner on the assignment. In addition, you **may not submit similar/identical solutions**.
- Provide a mathematical proof/computation steps to all questions, unless explicitly told not to. Solutions with no proof/explanations (even correct ones!) will be awarded no points.
- You may look up combinatorial inequalities and mathematical formulas online, but please do not look up the solutions. Most problems can be found in research papers/online; if you happen upon the solution before figuring it out yourself, mention this in the submission. **Unreferenced copies of online material will be considered plagiarism.**

1. (30 points) Provide a formal proof that the resolution inference algorithm is a sound and complete inference method. More formally, we are given a knowledge base KB represented in conjunctive normal form (and "AND of ORs"), and a query α, whose negation is also given in CNF form (i.e. it is another "AND of ORs").
   (a) (10 points) Prove that resolution in propositional logic is sound. 1z1
   (b) (20 points) Prove that resolution in propositional logic is complete, assuming that the query α is a single binary variable. You need to show that if applying the resolution algorithm to $KB \wedge \neg\alpha$ results in an empty resolvent then $KB \models \alpha$. You may assume that the knowledge base KB is internally consistent, i.e. it does not contain any contradictory statements, or statements that resolve to an empty resolvent $\perp$. Intuitively, it is the addition of $\neg\alpha$ that "causes" the empty resolvent.
   (c) (bonus points) Prove that resolution in propositional logic is complete for an arbitrary query α.

2. (20 points) Consider the adversarial regret minimization framework we studied in class. Provide a formal proof of the following claim. You may consult Chapter 4 "Learning, Regret Minimization, and Equilibria" from the Algorithmic Game Theory book for your solution, but use your own words and reasoning. We are given a two-player zero-sum game, where player 1's actions are $a_1, \dots, a_n$, and player 2's actions are $b_1, \dots, b_m$. The utility is given by an $n \times m$ matrix of values: $A \in \mathbb{R}^{n\times m}$, where $A_{i,j}$ is player 1's payoff if they play action i and player 2 plays action j. Since the game is zero-sum, player 2's utility is then $-A_{i,j}$. A strategy by player 1 is a distribution over $[n]$, and a strategy for player 2 is a distribution over $[m]$. Given a strategy by player 2, $\vec{q} = (q_1, \dots, q_m)$, we let $u_1(i, \vec{s}_2)$ be player 1's expected utility from playing action i, i.e.

$$u_1(i, \vec{q}) = \sum_{j=1}^{m} q_j A_{i,j}.$$

   Player 1's maximin value is

$$v_+ = \max_{i=1,\dots,n} \min_{\vec{q}\in\Delta([m])} u_1(i, \vec{q}).$$

Prove that if player 1 selects actions using a randomized policy with regret $\le R$, then their expected utility after T rounds is at least $v_+ - \frac{R}{T}$. See Theorem 4.9 in Chapter 4 of Algorithmic Game Theory.

3. (40 points) Bo (player 1) and Alexandra (player 2) are negotiating on how to split a cake. The negotiation works as follows. Each player i values the cake at a certain amount v_i. Each player can choose to either argue (A) or concede (C).

   - If both players choose to argue, they get half of the cake for a utility of $\frac{v_i}{2}$ for player i.
   - If one player argues while the other concedes, the player i who argues gets all the cake (for a utility of v_i) and the player who concedes gets a utility of 0.
   - If both players concede, each gets half the cake, with an additional utility of 2 for feeling better about not arguing. Thus, the utility of player i is $\frac{v_i}{2} + 2$ in this scenario.

   (a) (20 points) Suppose that Bo has one of two types. If Bo is of Type I_1, then his utility $v_1 = 1$. If he is of Type I_2 then his utility is $v_1 = M$, where $M > 4$. Alexandra has one type, and her utility is $v_2 = 2$. **First**, nature picks Bo's type (and, trivially, Alexandra's type): with probability $\frac{1}{2}$ Bo is of Type I_1 and with probability $\frac{1}{2}$ he is of Type I_2. Bo observes his type; Alexandra does not know Bo's type. **Second**, Bo chooses his action A or C; Alexandra does not observe Bo's choice. **Third**, Alexandra picks her action A or C and payoffs are determined. Model this as an extensive form game, mark the information sets, and compute the Bayes-Nash equilibrium of this game. Note that since payoffs are determined right after Alexandra picks her action, it does not matter very much whether Bo observes Alexandra's actions or not. For the purpose of this analysis you may assume that he observes her choice. You will receive 6 points for correctly writing the extensive form game tree + information sets, and 14 points for the equilibrium analysis.

   (b) (20 points) Suppose that Alexandra does not observe Bo's type, but does observe Bo's action, i.e. before Alexandra chooses whether to play A or C, she sees whether Bo chose A or C, but does not observe Bo's type. All other rules are the same. Model this as an extensive form game, mark the information sets, and compute the Bayes-Nash equilibrium in this case. You will receive 4 points for correctly writing the extensive form game tree + information sets, and 16 points for the equilibrium analysis.

   (c) (bonus points) Suppose that nature sets Bo to be of Type I_1 with probability p and of Type I_2 with probability $1 - p$. How does the equilibrium analysis change as a function of p ? Analyze the setting where Bo's choice of action is **not** observed by Alexandra, i.e. Part (a).

4. (10 points) In this question I would like you to reflect on what you have learned in class. Please describe **at least one** topic we have studied in class. How did this topic influence your understanding of AI? How will you apply this topic in your future endeavors (research, work, teaching etc.)? There are no "wrong" answers here, but trivial/one sentence answers will be marked as 0 (e.g., don't just write "I thought heuristic search was useful").